

END SEMESTER ASSESSMENT (ESA) - JULY - 2023**UE20CS353 - Compiler Design****Total Marks : 100.0**

1.a. Define Compiler.
What are the phases of the Compiler? Explain with a neat diagram.
Mention the input and output for each phase.

(10.0 Marks)

1.b. Explain with a diagram the interaction between Scanner and a Parser.

(5.0 Marks)

1.c. Explain 2 ways to handle reserved words and identifiers during the recognition of tokens.

(5.0 Marks)

2.a. Consider the following CFG, which has a set of terminals

$T = \{id, (,), [,], , ;\}$

$E \rightarrow id \mid id(A) \mid id[E]$

$A \rightarrow E \mid E ; A$

1. Left-factor this grammar. **(2 marks)**
2. Write FIRST and FOLLOW sets. **(2 marks)**
3. Construct an LL(1) parsing table for the left-factored grammar. **(3 marks)**
4. Show the operation of an LL(1) parser on the input string **id(id[id]; id)** **(3 marks)**

(10.0 Marks)

2.b. Consider following grammar production rules.

1: $E \rightarrow T + E$

2: $E \rightarrow T$

3: $T \rightarrow int * T$

4: $T \rightarrow int$

5: $T \rightarrow \epsilon$

Construct automaton and SLR parsing table.

(7.0 Marks)

2.c. Explain with a diagram, the model of an LR parser.

(3.0 Marks)

3.a. Define the following:

1. Synthesized attributes

2. L-attributed SDD

3. S-attributed SDD

(3.0 Marks)

3.b. The grammar to translate binary integer number into decimal is given below. Let synthesized attribute $B_N.val$ gives the value of decimal equivalent of binary number (eg. For input 11001, $B_N.val = 25$).

$B_N \rightarrow L$

$L \rightarrow L_1 B$

$L \rightarrow B$

$B \rightarrow 0$

$B \rightarrow 1$

Give a translation scheme for above grammar. **(5 marks)**

Draw annotated parse tree for '10111' **(3 marks)**

(8.0 Marks)

3.c. Explain parser stack implementation of the postfix SDT's with the help of the diagram. (5.0 Marks)

3.d. What is dependency graph? Explain with example.

(4.0 Marks)

4.a.

1. What is DAG?

2. Draw a syntax tree and DAG (Show stepwise common sub-expression identification) for the following expression. (Assume unambiguous grammar for arithmetic expressions)

$((x + y) - ((x + y) * (x - y))) + ((x + y) * (x - y))$

(7.0 Marks)

4.b. Provide Three-address code and its Quadruple, Triple and Indirect Triple Representation for $x = y[i] + z[j]$

(7.0 Marks)

4.c. Partition following sequence of three-address instructions into basic blocks. Also construct flow graph.

1. $i=1$
2. $j=1$
3. $t1=10*i$
4. $t2=t1+j$
5. $t3=8*t2$
6. $t4=t3-88$
7. $a[t4]=0.0$
8. $j=j+1$
9. If $j \leq 10$ goto (3)
10. $i=i+1$
11. If $i \leq 10$ goto (2)
12. $i=1$
13. $t5=i-1$
14. $t6=88*t5$
15. $a[t6]=1.0$
16. $i=i+1$
17. If $i \leq 10$ goto(13)

(6.0 Marks)

5.a. Explain the structure and each field of a general activation record with the diagram.

(6.0 Marks)

5.b. Generate target code for the following three-address statements, assuming static allocation for procedures. The code for p() and q() are kept at the memory locations 100 and 300 respectively. Activation records for p() and q() are kept at memory locations 400 and 600 respectively.

```
//Procedure p()
m=5
n=m*2
call q
halt
```

```
//Procedure q()
x=2*x
return
```

(5.0 Marks)

5.c. Explain calling sequence and division of tasks between caller and callee.

(4.0 Marks)

5.d. Generate target code sequence for the following basic block three address statements with register and address descriptors.

Assuming only two registers (R0 and R1) are available and all the variables are live on exit from the basic block. No temporary variables are used.

1. $x = y + z$
2. $z = x * x$
3. $y = z$
4. $x = y + z$

(5.0 Marks)